

IVORY & INK

"Truth in every shade."

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"IS THE GLOBAL WATER CYCLE AMPLIFYING CLIMATIC HAZARDS? "

By - Gaurika Chaudhary

For years, floods, droughts, and heatwaves have continued to dominate headlines around the world. Ever wondered why?

I'll tell you why! In the past few days, hundreds of bushfires have ignited in south-east Australia during an extreme heatwave. And communities in north Queensland have been lashed by heavy rain and flash flooding from ex-tropical Cyclone Koji. Behind these disasters is a deeper and less visible influence: **ongoing shifts in the global water cycle**. All of us know the process: water evaporates, falls as rain and snow, and ultimately evaporates again. These floods and fires are not simply isolated weather extremes, but signs of a water cycle that is being increasingly destabilised by global warming.

But why is the water cycle changing?

The global water cycle connects the atmosphere, land, oceans, and ice. Water evaporates from the land and seas, falls as rain and snow. This feeds glaciers, rivers, lakes, and groundwater, and finally either evaporates again or flows to the ocean. This cycle is driven by the energy from the sun. And as the planet warms, it is becoming more powerful and more erratic. Statistics show the last three years being the hottest on record, consequently leading to rapid global warming. High temperature accelerates evaporation from soil, vegetation, and inland waters, which ends up producing dry conditions more quickly than before. At the same time, warmer air can hold more moisture, which increases the potential for intense rainfall. Together, these processes intensify both floods and droughts.



Scientists describe these abrupt swings between dry and wet extremes as **"climate whiplash"** where ecosystems, infrastructure, and communities struggle to cope with wet and dry extremes. One example from 2025 is the **severe wildfires** in Spain and Portugal. Several other events in 2025 revealed how the water cycle is becoming more unstable.

In the Himalayas, a series of unprecedented **glacial lake floods** occurred within just a few months, followed by warm conditions. Meanwhile, a rare cyclone close to the equator took Indonesian and Malaysian communities by surprise. The increasing frequency of tropical cyclones in historically uncommon locations reflects how warming oceans and shifting atmospheric conditions are expanding the reach of water-related hazards.

Statistics show that water-related disasters in 2025 were associated with nearly 5,000 deaths, displaced around 8 million people, and caused economic losses exceeding US \$360 billion. In Europe, prolonged heatwaves were linked to many thousands of heat-related deaths. And in South and Southeast Asia, flooding displaced millions in countries such as the Philippines, Thailand, Malaysia, and Bangladesh.



This shows how different parts of the water cycle can influence ongoing global warming.

Now we know that preparedness really matters.

Early-warning systems and evacuation planning saved many lives in several major floods. However, severe disruption and economic damage still occurred where infrastructure had been designed for stable, historical conditions.

Conditions in the global water cycle at the end of 2025 point to greater drought risk in parts of the Mediterranean, the Horn of Africa, Brazil and Central Asia in 2026. On the other hand, wet conditions mean flood and landslide risks remain high in the Sahel, south of the Sahara Desert, southern Africa, northern Australia and much of Asia.

As climate instability continues, **the global water cycle is likely to become even more unpredictable.** Understanding how water moves through the climate system – and how quickly it can shift from one extreme to another – will help us reduce the impacts of future disasters.

Therefore, managing both heat and water extremes will be key to adapting to a rapidly warming world.

“WHEN THE LAND BENEATH US DISAPPEARS: INDIA’S SINKING DELTAS”

By - Anand Tiwari

River systems play a vital role in shaping society, supporting irrigation, drinking water, transportation, and power generation, and providing livelihoods to millions. That is why most major Indian cities like Delhi, Agra, and Ahmedabad are located along riverbanks. Serving as the nation's lifeline, rivers influence geography, culture, and the economy by sustaining ecosystems and fulfilling essential water needs.

River deltas occupy only 1% of global land area and are among the most vital landforms on Earth. Globally, deltas host an estimated 350–500 million people (representing 4–6% of the global population), including 10 of the 34 megacities of the world. Several of India's major river deltas, including the Ganga-Brahmaputra, Mahanadi, and Cauvery river deltas, serve as a livelihood for millions of Indians.

Despite their importance, these deltas are sinking at alarming rates because of excessive extraction of groundwater and rising sea levels, according to a new global study published in *Nature*. Climate change drives sea-level rise, and land sinks, causing deltas worldwide to face a "double burden" that pushes relative sea levels up faster than global averages, the researchers stated. Surveys by NGOs in India show that states such as Punjab, Haryana, and Rajasthan extract water at rates over 100% of their annual recharge. Although these regions lie upstream, their unsustainable water use disrupts river flows and sediment supply, which indirectly weakens delta systems downstream and causes land to sink. The science behind sinking deltas is actually quite straightforward and deeply worrying. When we pump out groundwater faster than nature can refill it, the soil underground starts to collapse. The tiny spaces between sediments shrink, the land sinks, and once that happens, there's no going back.

This isn't like floods or erosion, where land can sometimes recover. Satellite data now shows that large parts of India's major river deltas are sinking by several millimetres every year—sometimes even faster than sea levels are rising. That means floods become more frequent, coastlines wear away faster, and salty seawater creeps into freshwater sources, putting farms and drinking water at risk.

What is happening in India is part of a much bigger global problem. Across the world, deltas are sinking under similar pressures. Vietnam's Mekong Delta, despite being one of the planet's most important food-producing regions, is rapidly subsiding due to excessive groundwater use and dams upstream that block sediment. Egypt's Nile Delta faces the same threats. In the United States, decades of heavy human intervention have pushed large parts of the Mississippi Delta below sea level. These examples make one thing clear: delta degradation isn't inevitable, and it's largely the result of human decisions, weak policies and poor resource management. Unfortunately, India's response so far has fallen short. Groundwater laws are scattered, loosely enforced and often ignored. Development plans still treat rivers and deltas as fixed landscapes, rather than living systems that constantly change. Without controlling groundwater extraction, restoring natural sediment flows and planning with climate risks in mind, current efforts do little more than delay an inevitable crisis.

The sinking of India's deltas isn't some distant environmental issue; it's a slow-moving global emergency happening right beneath our feet. If we continue to look away, the consequences will be severe: food insecurity, loss of livelihoods and entire regions becoming unlivable. Saving these fragile landscapes will require strong political will, responsible water use, public awareness and a complete rethink of how development works with nature, not against it. Because once the land sinks for

good, rebuilding it may simply not be possible.

Degrading Ganga-Brahmaputra river delta
(Satellite view)
Source: Wikipedia



“THE TITANIC WILL NOT SINK NOW: A WAKE UP CALL”



By - Kirti Raj Jhala

A World Drowning in Waste: Once believed to be unsinkable, The Titanic sank because of human error,

overconfidence, and ignored warnings. Today, the world sails on a far more disturbing reality. Surrounded by floating plastic, garbage, and overflowing waste, One might ironically say, "The Titanic will

not sink now." Not because humanity has become wiser

- but because waste has replaced the ocean itself. Floating garbage gives the illusion of safety, but beneath it lies poisoned water and collapsing ecosystems. This is not

survival; this is pollution redefining nature.

Globally, the world generates over 2.2 billion tonnes of waste every year, and waste is growing faster than our solutions. What once was water and ice is now plastic and refuse. Waste Choking the Oceans:

Every year, nearly 8 million tonnes of plastic are dumped into the oceans. Plastic bottles take up to 450 years to decompose, breaking down into microplastics that enter marine life and human food chains. Scientists warn that by 2050, there may be more plastic than fish in the oceans (by weight) if current trends continue. Over one million seabirds and 100,000 Marine animals die annually due to plastic ingestion and entanglement. If garbage keeps ships afloat, it is because the ocean beneath is suffocating.

Lead, Mercury, and the Silent Poisoning of Life: Waste today is not just visible - it is toxic. Electronic and industrial waste releases lead, mercury, and other heavy metals into soil and water. These toxins damage the nervous system, contaminate food chains, and threaten future generations. The world produces over 50 million tonnes of e-waste each year, yet only 20% is recycled properly. The rest spreads silently through ecosystems, proving that waste never truly disappears.

Technological advancement without responsibility has turned progress into a threat.

India's waste reality : A global concern

India faces a serious waste and pollution problem. Rapid urbanisation, poor waste segregation, and overflowing landfills have created severe environmental and health challenges. The impact has drawn international attention. In recent years, foreign badminton players reportedly refused to participate in tournaments in India, citing unhygienic conditions and severe pollution. This highlights that waste is no longer only an environmental issue, it affects public health, global reputation, and international engagement.

A Lesson from Sweden: While many nations struggle, Sweden recycles or recovers nearly 99% of its waste through advanced recycling and waste-to-energy systems. Landfills are minimal, cities are clean, and waste is treated as a resource rather than a burden. Sweden proves that the Titanic can still be saved - if the course is changed now. Ignoring Waste Segregation Is Like Sailing The Titanic Without a Map Waste exists in many forms: **Plastic waste, Organic waste, Electronic waste, and Hazardous waste.** Without segregation, recycling fails and pollution becomes unavoidable. Sailing without waste management is like navigating the Titanic without direction-reckless and dangerous.

Waste: A Hidden Driver of Climate Change

Landfills emit methane, a greenhouse gas 25 times more powerful than carbon dioxide. Burning waste further increases emissions, accelerating climate change. Waste is a silent but powerful driver of global warming, intensifying floods, heatwaves, and extreme weather events.

The Future Crew of the Ship Is Watching Us:

Youth are not merely passengers-they are the future crew. Through awareness, innovation, responsible -



consumption, and action, young people can steer the planet toward safety. The choices made today will determine whether the ship survives tomorrow.

The Final Warning:

The Titanic did not sink because the iceberg was invisible - it sank because warnings were ignored. Today, waste is our iceberg: visible, expanding, and ignored. Floating garbage may keep objects above water, but it signals collapse beneath the surface. If garbage keeps us afloat, we are already sinking. The Titanic will not sink now, but the planet might, unless we change course.

“THE DILEMMA OF THE HILLS OF ARAVALI ”



By - Nitya Pareek

“The Aravallis don’t just stand as mountains – they stand as guardians of breath, balance, and Bharat’s future. When justice protects nature, it protects us all.”

It acts as a biological regulator, enabling species movement, genetic exchange, and long-term landscape-level biodiversity resilience.

The **ARAVALLI RANGE** is one of the world’s oldest fold mountain systems spread in the areas of Gujarat, Rajasthan, Haryana and Delhi. It acts as a barrier that prevents the expansion of the Thar Desert into the Indo Gangetic plains. It protects the Delhi-NCR region from intense dust storms originating from the desert.

In the latest events, the Supreme Court passed an order on the Aravallis, which helps the mining industries, where it was suggested that fertile land or soil up to the height of 100 m will be protected and called the Aravalli range, but below 100 m it can be mined and disturbed.

The illegal mining has been practised for decades, which pollutes the air and water particular region and even after all this, the Supreme Court ruled in favour of the miners.

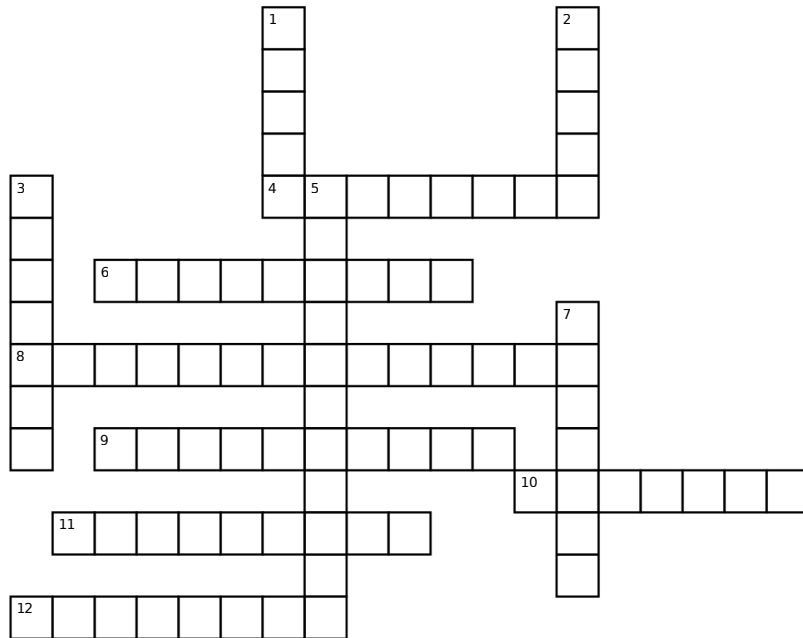
Then, in November 2023, due to several reasons and unprecedented protests Supreme Court took back its decision regarding the definition of the Aravalli hills and their ranges. The reason was due to the importance of demography, the



ecological continuity of ranges, which act as a green barrier against desertification. Also, the ruling resulted in widespread protests in Haryana and Rajasthan, with environmentalists warning that it could facilitate unregulated mining and cause irreversible damage to the ecosystem.

The long term sustainability of the Aravallis depends on balancing ecological preservation with responsible, sustainable development.

Thus, now it is time for us to be conscious of what’s happening around us and take action.

CROSSWORD PUZZLE**Down:**

1. Triangular landform formed by river deposition at its mouth.
2. Gas in the stratosphere that absorbs harmful ultraviolet radiation.
3. Day-to-day condition of the atmosphere.
5. The rigid outer layer of the Earth.
7. Low-pressure system with inward spiralling winds.

Across:

4. Height of a place above mean sea level.
6. Movement of people from one place to another.
8. Branch of geography dealing with population and human activities.
9. Envelope of gases surrounding the Earth.
10. Seasonal reversal of winds affecting the Indian subcontinent.
11. Zone where land, water and air interact to support life.
12. Imaginary lines running parallel to the equator.

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NOTE:

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